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RESEARCH INTERESTS

Self-supervised representation learning and **world models**, especially joint-embedding predictive architectures (JEPA) that learn by predicting in latent space; medical image analysis in low-label regimes; training-free guidance of text-to-image diffusion models.

EDUCATION

University of Dhaka

B.Sc. in Robotics and Mechatronics Engineering • CGPA: 3.20/4.00

Dhaka, Bangladesh

Expected Oct 2026

Relevant Coursework: Machine Learning, Artificial Intelligence, Intelligent Systems and Robotics, Digital Image Processing and Robot Vision, Digital Signal Processing, Linear Algebra, Statistics for Engineers.

PUBLICATIONS & MANUSCRIPTS

- **Mind the Gap: A Controlled Evaluation of Slice Separation and Masking in Self-Supervised CT Pretraining.** First author. Under review (double-blind).
- **GenMem: Auditing Predicted-Memorability Guidance in Text-to-Image Diffusion.** Undergraduate thesis; *manuscript in preparation*.

RESEARCH EXPERIENCE

Slice Separation and Masking in Self-Supervised CT Pretraining

Jan 2026 – Present

First-author study, University of Dhaka

[ajmainapon/SSLR](#)

- Adapted **I-JEPA** to 2.5D CT slice pairs (ViT-B/16 context encoder, EMA target encoder) on **1,082 unlabeled TotalSegmentator volumes**; a controlled 2×2 study over context–target slice gap and masking geometry showed slice separation dominates: the narrow-gap A2 backbone improves linear-probe Dice by **53–202%** over the wide-gap I-JEPA-CT baseline, and the two factors interact non-additively.
- A2 outperforms random init by **+102–118%** under linear probing and **+13.0/8.5/2.8%** under end-to-end fine-tuning at $N=20/50/100$ labels (Welch t : $p=0.017$, $p<0.001$); reaches **67%** of supervised **nnU-Net** at $N=50$ (liver **90%**); reduces seed variance **1.3–2.2×**.
- Validated **cross-dataset** transfer to AMOS22-CT (**+158–260%** over random init) and **cross-modality** CT→MR with no MR pretraining (**+41–48%**); a matched 3D **SwinUNETR** comparison reproduces the same label-dependent decline in fine-tuning gain.

Senior Thesis — Predicted-Memorability Guidance in Diffusion

Jan 2026 – Present

Advisors: Dr. Md. Mehedi Hasan, Dr. Sejuti Rahman, University of Dhaka

[ajmainapon/GenMem](#)

- Developed **GenMem**, a training-free guidance method that differentiates a pretrained memorability predictor (**AMNet**) through the **DDIM** sampling trajectory of **Stable Diffusion**, leaving all model weights frozen.
- On **DrawBench** ($n=200$): memorability scored by gradient-held-out predictors rose **+3.42%** (ResMem) and **+2.72%** (ViTMem), both $p < 10^{-8}$, with no significant CLIP-fidelity loss; replicated on **COCO** ($n=200$) with equal or larger effect sizes.
- Tested causality via reverse-guidance symmetry and a null random-critic control; characterized proxy over-optimization in SDXL and mapped a three-regime dose–response curve across an 8-point guidance-weight sweep.

Undergraduate Research Assistant — Stomatal Phenotyping

2024 – 2025

Funded project, University of Dhaka; paper submitted to *Frontiers in Plant Science*

[ajmainapon/StomataAI](#)

- Built a **506-image two-class instance-segmentation benchmark** (6,605 guard-cell-complex and pore annotations) from chickpea leaf microscopy, with locked image-level train/val/test splits (359/99/48).
- Benchmarked **YOLOv8s-seg**, **YOLO26s-seg**, and **Mask R-CNN** on identical splits: YOLO26s-seg led mask **mAP@0.50 (0.638)**, Mask R-CNN led mask **mAP@0.50:0.95 (0.428)** on the held-out test set.
- Automated stomatal density and pore-index extraction from predicted masks, validated against manual counts ($p < 0.05$); used PCA and t -tests to relate stomatal and morpho-physiological traits to drought stress in *C. arietinum*.

HONORS & AWARDS

- **Champion, Insight 2.0 Datathon 2026** (79 teams), IASDS Students' Club, University of Dhaka: lung-cancer survival prediction from SEER registry data (**0.880 weighted F1**).
Sep 2026

SKILLS

Programming Python, C++, C

Frameworks PyTorch, timm, MONAI, Hugging Face, torchvision, Ultralytics, Scikit-learn, OpenCV, Nibabel, CUDA

Methods Self-supervised pretraining, diffusion guidance, instance segmentation, controlled ablations, statistical testing

LEADERSHIP & SERVICE

- **President**, Robotics & Mechatronics Student Club, University of Dhaka
- **Program Coordinator**, IEEE Robotics & Automation Society, University of Dhaka

Oct 2025 – Present

Mar 2024 – Apr 2025